A Study of Factors Affecting Process-induced Warpage Behavior of Flip Chip Package on Package

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***Abstract*—**This study attempts to investigate the crucial factors affecting the warpage behavior of flip chip package on package (fcPoP) during fabrication. To characterize the process-induced warpage, a process-dependent simulation methodology that integrates ANSYS element death/birth technique and finite element analysis (FEA) is introduced. For modeling simplicity, the core and coreless substrates are transformed into an equivalent homogenized material, and their effective thermal-mechanical properties are determined using a proposed effective model based on FEA and trace mapping feature in ANSYS. The predicted effective mechanical properties are compared with those obtained by the rule-of-mixture analytical model and also the detailed FEA as a benchmark. Besides, the predicted process-induced warpages during the fabrication process are validated against experimental data.

Keywords*——Flip Chip Package on Package, Warpage, Process Simulation, Finite Element Method, Effective Modeling*

I. Introduction

Recently, three-dimensional (3D)/stacked packaging become mainstream due to its heterogeneous integration performance. Among the 3D/stacked packaging technologies, package-on-package (PoP) [1], including fan-out PoP [2] and 3D PoP with flip chip (FC) interconnect (briefly termed fcPoP) (see Fig. 1), has become one of the most dominant and competitive packaging technologies today. As compared to flip chip chip-scale package (fcCSP), fcPoP can provide lower power consumption, smaller form factor, greater performance, higher bandwidth. However, the packaging technology still faces many significant technical challenges. For example, the particular structure of the fcPoP shows a considerable mismatch of the coefficient of thermal expansion (CTE) among materials, which may cause severe warpage during fabrication process, thereby challenging the subsequent process operations. It is essential to have a good understanding and control of its process-induced warpage behavior. It is, however, important to note that coreless substrate (alternatively termed SOC substrate) tends to undergo much more significant warpage than traditional substrates because of a lack of rigid core material to support the package [3].

II. . Finite Element Modeling

A. Effective modeling

Figure 2 schematically reveals the cross sectional view of the core substrate and coreless substrate. The substrates generally require extensive and tedious modeling because they comprise a composite structure with Cu trace. To ease the modeling challenges, the substrates are approximated in this work as an equivalent homogeneous medium, and their effective mechanical properties are evaluated using a proposed effective approach that makes use of the powerful trace mapping feature in ANSYS together with FEA. This approach is hereinafter termed FEA-based effective approach. The effective model can not only greatly reduce the modeling effort, but also significantly enhance computational efficiency while still maintaining solution accuracy.

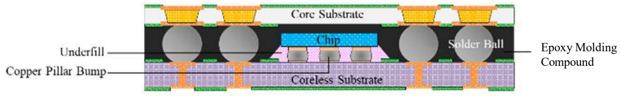
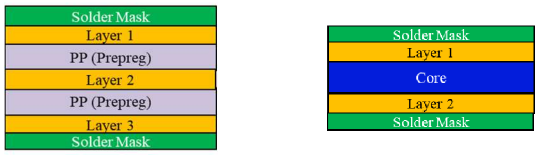


Fig. 1 Schematic of the fcPoP packaging technology



(a) Coreless substrate (b) Core substrate

Fig. 2 Schematic of coreless and core substrates in cross-sectional view

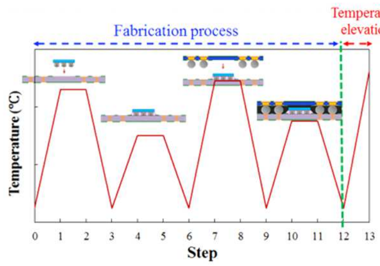


Fig. 3 Fabrication process and temperature elevation

B. Process Warpage Modeling

The fabrication process steps together with thermal loads are displayed in Fig. 3. The process modeling integrates thermal and mechanical FEAs and element death-birth feature in ANSYS together with the constructed effective models of the core and coreless substrates. The FEA takes into account the viscoelastic behavior and cure shrinkage of the epoxy molding compound (EMC). The study starts from experimentally characterizing the temperature-dependent thermal- mechanical properties used in the fcPoP, using thermal mechanical analyzer (TMA) and dynamic mechanical analyzer (DMA).

III. Results and Discussion

A. Verification of the proposed effective model

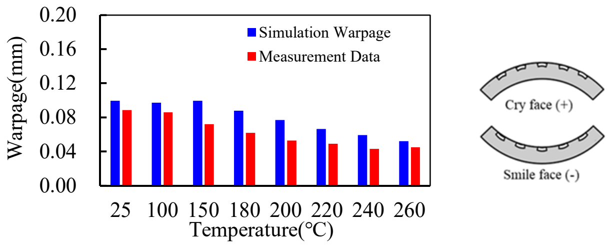
For the feasibility test of the proposed effective approach, the coreless substrate with a silicon chip subjected to a temperature swing load (from the room temperature to 260 °C) is simulated. The predicted effective thermal-mechanical properties using the proposed FEA-based effective approach are compared against the rule of mixture (ROM) technique. For comparison, a detailed FEA is also conducted as a benchmark. The effective approaches and analysis results are displayed in Table 1, which shows that the presently proposed FEA-based effective approach has a good agreement with the benchmark. This indicates how the proposed FEA-based effective approach can be used as an effective means for characterizing the equivalent thermal-mechanical properties

2021 International Conference on Electronics Packaging (ICEP) | 978-4-9911-9111-4/21/$31.00 ©2021 | DOI: 10.23919/ICEP51988.2021.9451980

of a composite structure with high geometry complexity and multi-scale essence.

Table I Comparison of the predicted warpages using the ROM/energy-based analytical estimate and FEA-based effective approaches, and also the warpage obtained by the detailed FEA as a benchmark.

Young's Modulus	CTE	Warpage (mm) (Effective Modeling)	Warpage (mm) (Benchmark)	Difference
ROM	**Analytical Estimate**	0.552	0.524	5.45%
FEM	**FEM**	0.525	0.524	0.18%



(a) Warpage during temperature elevation (b) Warpage orientation

Fig. 4 The modeled and experimental process-induced warpage and definition of warpage orientation

B. Process-induced warpage

During the temperature elevation from 30 °C to 260 °C, the modeled results are shown in Fig. 4, where the warpage orientation is defined in Fig. 4(b). The analysis result shows the process-induced warpage decreases with temperature as the temperature increases from 30 °C to 260 °C, and the fcPoP warps in a concave shape. Similar result is also found in the experimental data. Additionally, there is a very consistent warpage result between the simulation and experiment. The difference between them is about 14%, fairly suggesting the effectiveness of the proposed process warpage modeling. The nontrivial difference is probably due to measurement uncertainty in material mechanical properties.

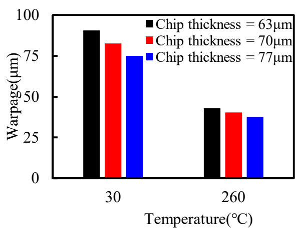


Fig. 5 The effect of chip thickness

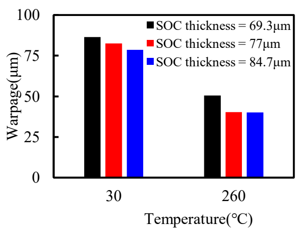


Fig. 6 The effect of SOC substrate thickness

C. Parametric study

The effects of some crucial parameters on the process-induced warpages at Step 12 and Step 13. Figure 5 reveals that the warpages at Step 12 and Step 13 decrease. Basically, the increase of the chip thickness would enhance the rigidity of the package, thereby leading to a reduced warpage. The effect of the thickness of the SOC substrate on the warpages is shown in Fig. 6. The figure shows that similar to the effect of chip thickness, an increasing SOC substrate thickness would also reduce these two warpages. The reason is also similar to that addressed in the effect of chip thickness. Besides, The influence of EMC thickness is investigated and the results are displayed in Fig. 7. Figure 7 demonstrates that a thicker EMC would greatly lessen the warpage at Step 12, probably due to the increase in the elastic rigidity. In contrast, a different result is found on the warpage at Step 13, where there is a turning point: the warpage decreases as the EMC thickness increases from 117 μm to 130 μm, and then increases as it increases from 130 μm to 143 μm.

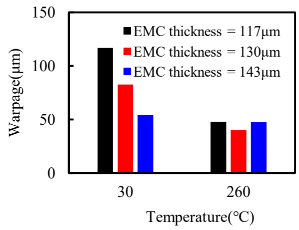


Fig. 7 The effect of EMC thickness

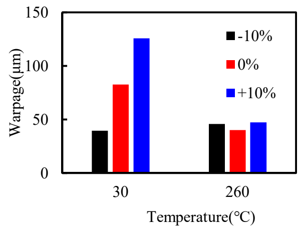


Fig. 8 The effect of SOC substrate CTE

At last, the influence of the CTE of the SOC substrate is shown in Fig. 8. It is found that the warpage at Step 12 increases with an increasing SOC substrate CTE, mainly owing to the increase in the CTE mismatch between the core and coreless substrates. On the other hand, there is an inflection point in the effect of the SOC substrate CTE on the warpage at Step 13, similar to the effect of SOC substrate thickness, where the warpage initially decreases and then increases with an increasing SOC substrate CTE.

IV. Conclusions

The study investigates the effects of some crucial parameters on the process-induced warpage of the fcPoP. For modeling simplicity, the core and coreless substrates are modeled as an equivalent homogenized medium using the proposed effective approach. It turns out that the proposed FEA-based effective approach can greatly upgrade computational efficiency while still maintaining solution accuracy. The effectiveness of the proposed process warpage modeling is also demonstrated by comparing with the experiment. Results show that there is a good consistency in the predicted warpage between the presently proposed process warpage simulation and the experiment. The parametric study shows that an increasing chip thickness, SOC substrate thickness and EMC thickness and a decreasing SOC substrate CTE can lessen the warpage at Step 12.

V. Acknowledgment

The work is partially supported by Ministry of Science and Technology, Taiwan, ROC, under grants MOST 107- 2221-E-035-022-MY3. The authors also thank the National Center for High-Performance Computing (NCHC) for computational resources support.

VI. References

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